

Dirac spinor normalization

A widely-used convention is

$$u^\dagger u = 2E$$

where E is relativistic energy

$$E = \sqrt{\vec{p} \cdot \vec{p} + m^2}$$

For example, for a spin-up electron

$$u = \sqrt{E + m} \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E + m} \\ \frac{p_x + ip_y}{E + m} \end{pmatrix} \quad u^\dagger u = 2E \quad (1)$$

For $\vec{p} = 0$ we have $\sqrt{E + m} = \sqrt{2m}$ hence

$$u = \sqrt{2m} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad u^\dagger u = 2m = 2E$$

Eigenmath script